

Landing Page A/B Test — Conversion and Revenue Validation

TripleTen Data Analytics Bootcamp · 28-day homepage experiment · 40,000 users

Statistical validation of an A/B experiment on the homepage of an e-commerce platform, run to decide whether version B should replace version A. Each comparison is matched to a test its assumptions justify — Levene before the t-test, a z-test for proportions, chi-square on counts for the segment questions — and every significant result is reported with its effect size, because at n = 40,000 significance is cheap and relevance is not. Tests that would strengthen the analysis but were not run are listed, not implied.

Business Question

Business Question:

Which version of the homepage — A (control) or B (variant) — should be implemented?

- Is there a significant difference in average spend per converted user between versions?
- Which version produces the higher conversion rate, and by how much?
- Does conversion depend on the traffic source a visitor arrives through?
- Does user type (new vs returning) influence conversion?
- How large is each difference, and is it large enough to act on?

Methodology

- Load and validate the data:** 40,000 rows x 9 columns checked for missing values, duplicate rows and repeated user IDs; category labels and column names standardized; group composition compared across region, device, channel and user type.
- Compare spending per converted user:** test the equality-of-variances assumption with Levene, then apply Welch's t-test, and report the 95% confidence interval and Cohen's d alongside the p-value.
- Review the extreme spending values** with the IQR rule inside each version, confirm they are genuine purchases rather than errors, and re-run the comparison without them as a sensitivity check before deciding whether to keep them.
- Compare conversion rates** between versions with a two-proportion z-test, reporting the difference in percentage points and in relative terms.
- Test the segment questions** with chi-square on contingency tables of counts — traffic source and user type — validating expected frequencies.
- Quantify every significant association** with Cramér's V, so that statistical significance is never read as practical relevance at n = 40,000.
- Visualize volume against effectiveness** for both categorical variables: absolute counts beside relative frequencies, since a large channel and an effective channel are different things.

```
# Data quality and consistency checks before any comparison
run_initial_checks(df)          # shape, dtypes, nulls, duplicates, repeated user_id

# Spending: the assumption check decides which t-test is valid
l_stat, p_value_var = levene(spending_A, spending_B)      # 6.9e-08 -> unequal variances
t_result = ttest_ind(spending_A, spending_B, equal_var=False)  # Welch

# Effect size and interval: the p-value says the difference exists, not how large it is
ci_low, ci_high = t_result.confidence_interval(confidence_level=0.95)
pooled_sd = np.sqrt((spending_A.var(ddof=1) + spending_B.var(ddof=1)) / 2)
cohens_d = (spending_B.mean() - spending_A.mean()) / pooled_sd

# Outliers: IQR rule inside each version, then a sensitivity re-run of the same test
q1, q3 = series.quantile([0.25, 0.75]); iqr = q3 - q1
low, high = q1 - 1.5 * iqr, q3 + 1.5 * iqr
t_trimmed = ttest_ind(trimmed_A, trimmed_B, equal_var=False)  # +7.61 vs +7.66 -> kept

# Conversion rate: two-proportion z-test
stat, pval = proportions_ztest(counts, nobs)

# Segments: chi-square on COUNTS (never on a percentage table)
contingency_table = pd.crosstab(df["traffic_source"], df["converted"])
chi2_stat, p_val, dof, expected = chi2_contingency(contingency_table)

# Effect size for the chi-square, on a 0-1 scale that does not grow with n
cramers_v = np.sqrt(chi2_stat / (len(df) * (min(contingency_table.shape) - 1)))
```

Python / pandas / scipy / statsmodels sample (data checks, test selection and effect sizes): representative, not complete — the final notebook combines this with the exploration, contingency tables and visualization steps.

Key Finding (Context → Finding → Implication)

Context

A 28-day homepage experiment exposed 40,000 users, split 19,982 / 20,018.

No missing values, no duplicate rows and no repeated users; the two groups are built from comparable audiences on region, device, channel and user type.

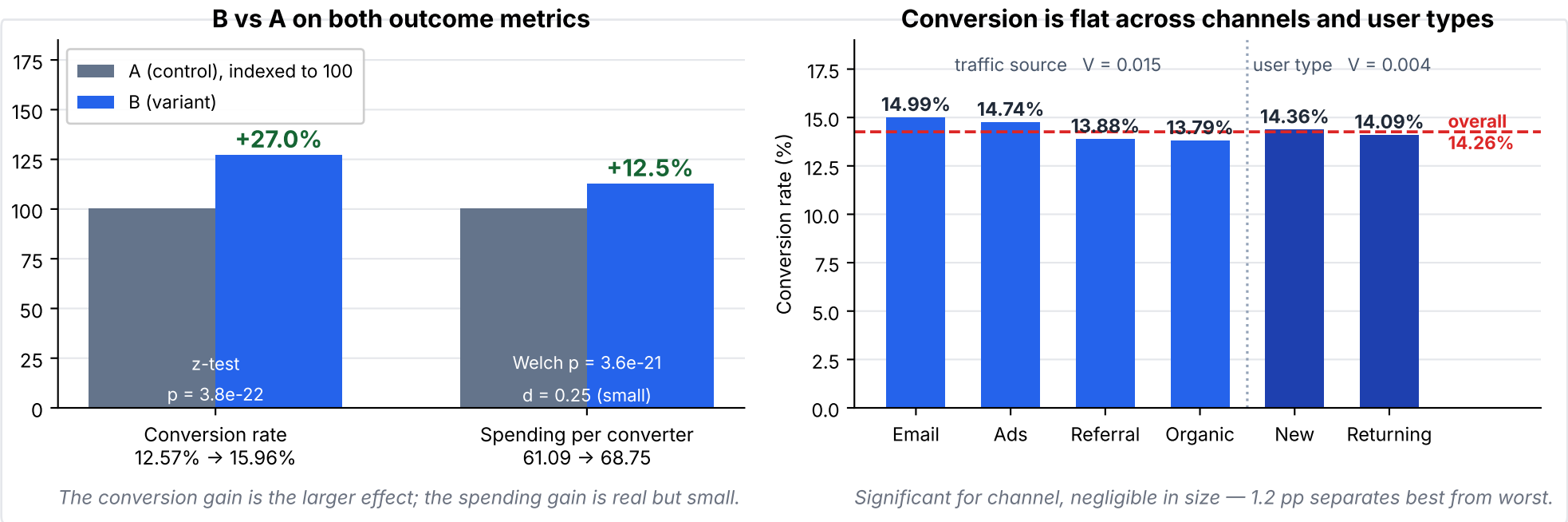
Finding

B converts 15.96% vs 12.57% — +3.38 pp, +26.9%, z-test p = 3.8e-22. Its converters also spend more: 68.75 vs 61.09, +12.5%, 95% CI [+6.08, +9.24], but Cohen's d = 0.25 marks that second effect as small. Neither segment variable matters.

Implication

Implement B, and treat the conversion gain as the prize rather than the spending difference. Do not segment the decision by channel or user type. Validate the allocation and run the tests listed under Next Analyses before the roll-out.

Visualization



Main Findings

Result	Test	Value	Read
Conversion rate — A vs B 12.57% → 15.96%, p = 3.8e-22 — about 680 extra converters per 20,000 visitors	Two-proportion z-test	+3.38 pp (+26.9%)	DECISIVE
Spending per converted user 61.09 → 68.75, p = 3.6e-21, 95% CI [+6.08, +9.24]	Welch's t-test	+7.66 (+12.5%)	REAL
Size of the spending effect The two spending distributions overlap substantially — a modest shift, not a new kind of buyer	Cohen's d	d = 0.25	SMALL
Extreme spending values Genuine purchases, evenly split; without them the difference is +7.61 and d rises to 0.30	IQR rule + sensitivity re-test	2.83% A · 2.79% B	KEPT
Variance assumption Unequal variances rule out Student's t-test and select the Welch version	Levene's test	p = 6.9e-08	WELCH REQUIRED
Traffic source ↔ conversion Email 14.99% · Ads 14.74% · Referral 13.88% · Organic 13.79% — all expected frequencies valid	χ² on counts	χ² = 8.66, p = 0.034	SIGNIFICANT
Size of that association The whole spread across channels is 1.2 pp — no basis for reallocating budget	Cramér's V	V = 0.015	NEGLIGIBLE
User type ↔ conversion New 14.36% vs returning 14.09% — entirely compatible with random variation	χ² on counts + Cramér's V	p = 0.474, V = 0.004	NO ASSOCIATION

Significance vs relevance: the channel-to-conversion association is statistically significant ($\chi^2 = 8.66$, $p = 0.034$) and still worth nothing as a lever: Cramér's $V = 0.015$ and the entire spread across channels is 1.2 pp. At n = 40,000 a trivial difference clears any threshold; the effect size, not the p-value, is the decision-relevant quantity. The same logic qualifies the spending result, where $d = 0.25$.

Chi-square takes counts, not percentages: run on a table normalized to percentages the same test returns $\chi^2 = 0.089$, $p = 0.993$ — an apparently decisive "no association" that is an artifact of the table total being 400 instead of 40,000. The statistic scales with the table total, so the table has to carry the raw counts.

Next analyses, not yet run: sample-ratio-mismatch and covariate-balance tests to validate the allocation formally; Shapiro-Wilk and Mann-Whitney U to confirm the spending result survives the skew; a confidence interval for the conversion difference; Breslow-Day plus Holm-corrected segment tests to ask whether B's advantage is uniform; a bootstrap on revenue per exposed user, which combines both effects; and an analysis of the upper tail, where ~7% of revenue sits with 2.8% of buyers. The notebook lists each one with the gap it closes.